

PROGESTERONE-MODULATION OF ESTROGEN ACTION: RAPID DOWN REGULATION OF
NUCLEAR ACCEPTOR SITES FOR THE ESTROGEN RECEPTOR

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SUMMARY

Our previous studies demonstrated that progesterone down regulates the occupied form of nuclear estrogen receptor (Re). Using the density shift method, we discovered that progestins stimulate the turnover of nuclear Re within 3 h of treatment, and Re synthesis is suppressed subsequently. Thus, the primary site of progestin action in down-regulating Re is the stimulation of nuclear Re turnover followed by the inhibition of Re replenishment. A major breakthrough in our understanding of how progestin controls Re turnover was made by studying nuclear acceptor sites for Re that were found to decrease markedly within 2 h of progestin treatment. These and other results indicate that progestin induces a factor called the Re regulatory factor (ReRF) which acts to block nuclear Re acceptor sites, and this in turn decreases nuclear Re retention on chromatin acceptor sites, leading to an enhanced turnover (or processing) of nuclear Re.

INTRODUCTION

The ability of progesterone (P) to regulate estrogen (E) action in the mammalian uterus is crucial in establishing and maintaining reproductive competency. During pregnancy, P acts by way of a specific receptor (Rp) to induce a P-dependent RNA and protein complement while decreasing E-dependent gene transcription. Recent work from our laboratory (1,2) indicates that P modulation of E-dependent gene transcription involves a specific P-dependent loss of nuclear estrogen receptor (Re). Other studies conducted using tumor cell lines which contain Rp and Re supply additional insight about E and P interactions (3,4). The existence of Rp (and to a lesser extent, Re) in these cells correlates with the efficacy of steroid hormone therapies in tumor regression. Certainly, the modulation of E action by P plays a vital role in both normal and abnormal reproductive tissues (5,6). The purpose of this chapter is to review recent work that has revealed a novel mechanism by which P controls E action at the level of the target cell nucleus.

In order to understand how P action is mediated, we must first review Re dynamics (Fig. 1). Estrogen action in target tissues occurs when E

enters the cell, binds to its high affinity, ligand specific, low capacity receptor protein (Re), which is loosely associated with various nuclear components (7). After E binds to Re, the ReE complex is rapidly modified or "activated" (7-9), changing its physicochemical properties with an increase in affinity for nuclear binding sites, called acceptor sites, which are believed to be composed of DNA and chromatin proteins (10). By convention, the loosely-bound ReE complex is referred to as cytosolic ReE, since it is found in the cytosolic fraction after tissue homogenization (8). The tightly-bound ReE complex is referred to as nuclear ReE since

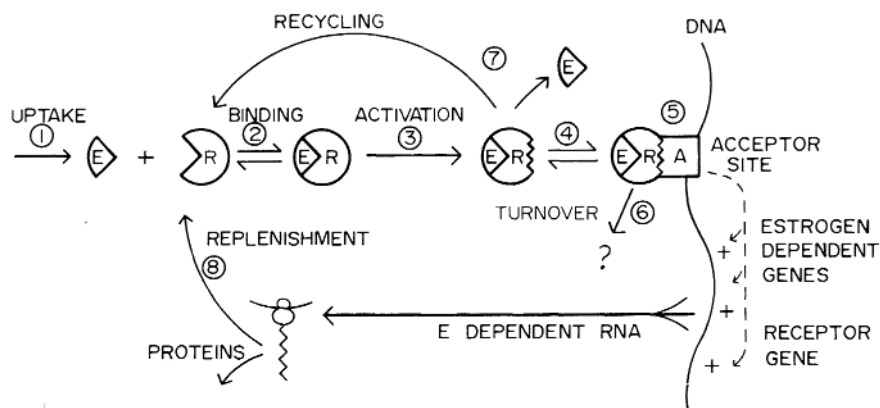


Fig. 1. Estrogen receptor dynamics in a uterine target cell. Steps are: (1) hormone uptake by the cell; (2) binding to unoccupied receptor; (3) receptor activation; (4) binding to acceptor sites; (5) receptor retention; (6) receptor processing; (7) receptor recycling; and (8) receptor replenishment. Abbreviations are: E, estradiol-17 β ; R, estrogen receptor; A, nuclear acceptor site. Adapted from Leavitt (2).

its removal from the nuclear fraction requires extraction with high salt concentration, pyridoxal 5'phosphate or thiocyanate (11). A similar scheme is believed to apply for P and its receptor, Rp (7,12). Retention of ReE by nuclear binding sites has been correlated with an increase in the transcription and translation of Rp, Re, oxytocin receptor proteins in the uterus, the egg white proteins in the chick oviduct, and a host of other E-induced proteins. Decrease in ReE retention is similarly correlated with decreased levels of these proteins (13-17).

METHODS

Our lab has been interested in steroid receptor regulation for several years (1,2,12,18), and the results that we will review in this chapter address the question, how does P control the Re system. We will first review pertinent physiological studies which directed us in terms of how to design experiments to address this question. The first line of evidence comes from the estrogen-dominated uterus that one finds during the hamster estrous cycle, and we will review the acute effects of P on Re in the

proestrous hamster uterus. The second line of evidence comes from the P-dominated myometrium, which relates to pregnancy maintenance and also the mechanism of parturition at the end of pregnancy. The third line of evidence has to do with recent studies we have done on Re synthesis and degradation in uterine decidual cells, as measured by the density-shift method. Finally, acceptor site studies, using methods adapted from Spelsberg et al. (10), have permitted us to identify a binding site in uterine chromatin which seems to qualify as an Re acceptor site. We describe for the first time a very rapid effect of P on the regulation of these acceptor sites in the proestrous hamster uterus.

RESULTS

In the four-day estrous cycle of the hamster (Fig. 2), there is a preovulatory surge of P secretion on the day of proestrus (day 4), as occurs in other rodent species such as the rat and mouse. As estrogen secretion increases during follicular development on days 2-4 of the cycle, there is an increase in uterine nuclear Re levels, and also an induction of Rp, reflecting the up regulation of receptors that one sees with E action in the uterus. Down regulation begins with preovulatory P secretion at mid-cycle. One sees a very rapid effect of P on the Rp system, and the net result in terms of Rp levels is the down regulation of Rp. There is also a loss of nuclear Re at mid-cycle, and we became interested in the cause of this down regulation. The possibilities were that this was a result of 1) estrogen withdrawal, 2) P action, or 3) a combination of 1 and 2. As detailed elsewhere (19), we were able to determine that the rapid decrease in the nuclear Re was really the result of P action, and not estrogen withdrawal. This observation prompted our interest in what P was doing at the level of the target cell nucleus. Thus, we designed experiments to determine which form of Re might be regulated and, using conventional receptor assays (20) as well as new assays developed to optimize the recovery of occupied Re (11), we found, as shown in Fig. 3 using the E-dominated uterus, that there was a very rapid and selective effect of P treatment on the occupied form of nuclear Re (21). This was of considerable interest because it suggested that P controlled the biologically important form of the receptor, i.e., nuclear occupied Re. Furthermore, in temporal studies, P had a very rapid effect on the occupied form of nuclear Re occurring within 2 to 4 h (20).

These and other studies (22) suggested to us that P was working by way of its receptor system perhaps to induce some RNA or protein which then either produced a new factor or activated a pre-existing factor which we called the estrogen receptor regulatory factor (ReRF). We first proposed that this factor might act to stimulate degradation of the occupied form of the nuclear Re (1). Since we were not able to prove that this factor acts directly on the receptor (2), an alternative hypothesis was that the factor may act on what the hormone receptor complex is associated with in the chromatin, i.e., the nuclear acceptor site.

We were also interested in receptor regulation during pregnancy where Re and Rp are chronically down regulated (23), and in particular, what happens when P is withdrawn, in terms of the recovery of the Re system and also estrogen-dependent proteins. In this situation, we used ovariectomized animals given Silastic pellets containing E and P, which maintain blood levels of E and P equivalent to what is present during normal pregnancy (23). Our plan was to maintain E levels at a steady state, so that the only hormone that changes at time zero, upon removal of the P pellet, is the fall in serum P. Within 4 to 8 h of P withdrawal, we see a significant recovery of nuclear Re and cytosol Re (23). Thus, under steady state E

exposure, the recovery of Re is accompanied by an increase in estrogen-dependent proteins such as Rp in the cytosol fraction and the oxytocin receptor in the membrane fraction. These observations are consistent with the idea that P down regulates the nuclear Re chronically, and in the withdrawal paradigm, we see a recovery not only of the occupied nuclear Re from extremely low levels within 4 h (Fig. 4), but we also see a signi-

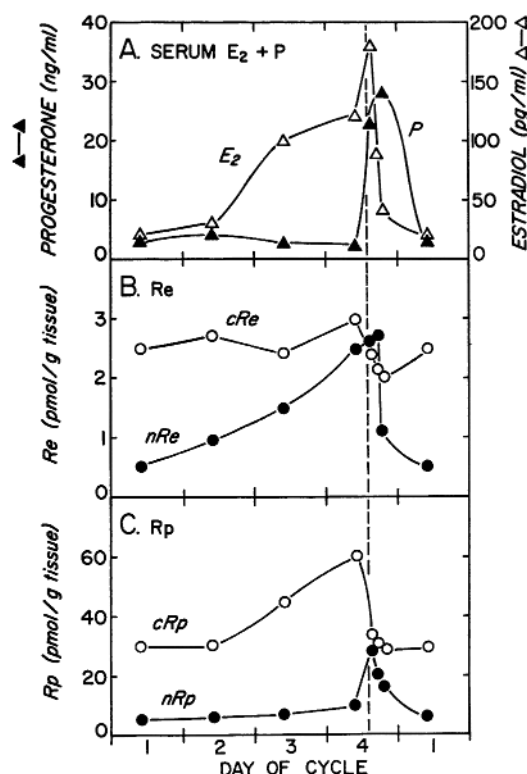


Fig. 2. Serum steroid-uterine receptor relationships during the hamster estrous cycle. The dashed vertical line indicates the time of the critical period for the ovulatory surge of gonadotropin. Abbreviations: c, cytosol; E_2 , estradiol-17 β ; n, nuclear; P, progesterone; Re, estrogen receptor; Rp, progesterone receptor. Adapted from Leavitt et al. (1).

ficant recovery of occupied cytosol Re (21). So the situation here may not be exactly equivalent to a reversal of the acute down regulation that we see in response to P action in the estrogen-primed uterus, but a similar phenomenon seems to be operative. This provides further support for the ReRF hypothesis that P induces something that controls the occupied form of Re perhaps by regulation of Re retention on the nuclear acceptor sites.

Clearly, the hormone withdrawal studies show that this effect is rapidly reversed when P is removed from the target cell.

We have learned how to culture hamster decidual cells (24), and with these cells, we can study up and down regulation of receptors because the cells will grow either in the presence or absence of P under our culture conditions (25). If we grow them in the presence of P, and then withdraw P at time zero, we find that both Rp and Re recover within 8 to 16 h (25). Next, we wanted to use these cells to study the effect of P on Re dynamics,

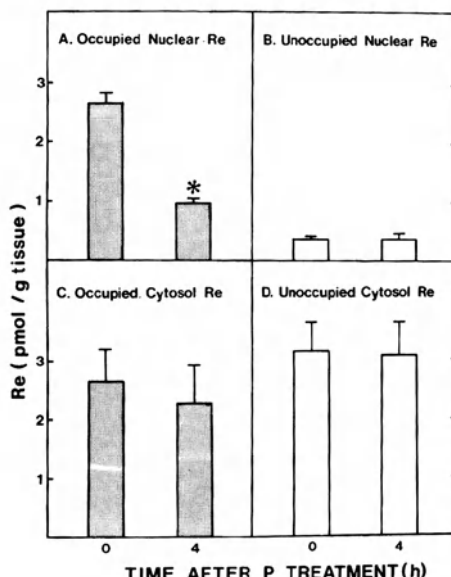


Fig. 3. Effect of progesterone (P) treatment on occupied and unoccupied estrogen receptor (Re) in proestrous hamster uterus. P (5mg/100 g body wt) was injected subcutaneously at 0900 h on cycle day 4. Time 0 controls were injected with 0.3 ml corn oil vehicle. *Significantly different from control (time 0). From Leavitt and Okulicz (21).

and this was accomplished using the density-shift method to measure Re synthesis and degradation. Fig. 5 outlines how this experiment is conducted. First, the cells are grown in the absence of hormones. Then Re is charged with unlabeled E_{13} and at time zero we expose the cells to dense amino acids (containing 2H , ^{13}C , ^{15}N). The dense amino acids are incorporated into the proteins being synthesized under these conditions. Since the Re is one of the proteins that is synthesized, it incorporates the heavy amino acids,

producing the so-called "heavy" form of Re which we identify by sucrose gradient centrifugation. The Re of normal density is the receptor that was present at the start of the experiment, and the decay of the normal density form of the Re determines how fast the receptor protein is being degraded. The rate of appearance of the heavy-density form of Re is a measure of how fast the receptor is being synthesized. Experiments were

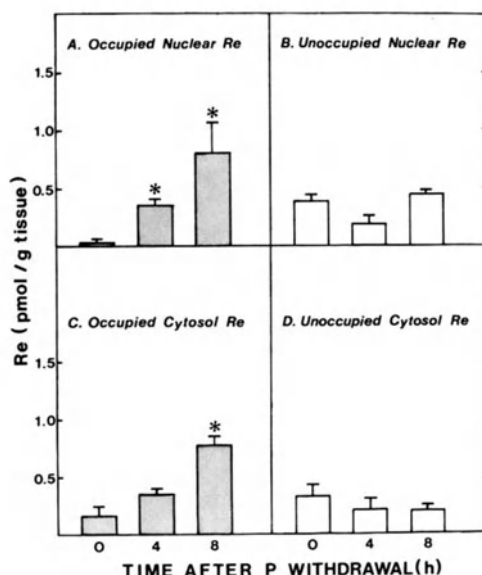


Fig. 4. Effect of progesterone (P) withdrawal on occupied and unoccupied estrogen receptor (Re) in myometrium of deciduized hamster uterus. Pseudopregnant (PS) hamsters were ovariectomized and given subcutaneous Silastic implants of estradiol and P. Decidualization was induced on PS day 4 and P withdrawal was initiated in morning of PS day 7 at time 0. *Significantly different from time 0, $P < 0.05$. From Leavitt and Okulicz (21).

done where the cells were exposed to E or E + P during incubation with dense amino acids. Fig. 6 depicts the gradient results. The Re of normal density is labeled with [125 I]-estradiol, and it sediments at 4S under these conditions. At 1 h, one begins to see a heavy-density form of Re at 6S, and with time, the 6S peak increases as the heavy amino acids are incorporated into the newly synthesized receptor. The 4S peak decreases with

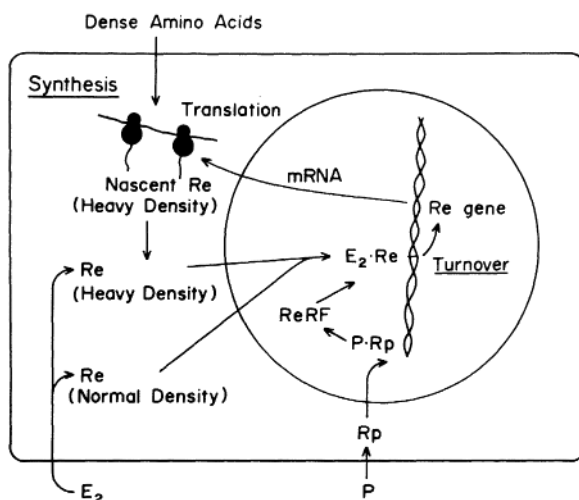


Fig. 5. Outline of density-shift experiment using decidual cell cultures. Decidual cells are first grown in the absence of hormone (25), and then the cells are incubated with unlabeled estradiol (E_2) to charge Re. At time zero, the cells are incubated in medium enriched with dense amino acids containing the heavy isotopes [2H , ^{13}C , ^{15}N]. Two hormonal regimens were compared: E_2 alone versus E_2 plus P. At various times of incubation, the nuclear Re was extracted, labeled by exchange with [^{125}I]-estradiol, and the Re of normal density (4S) and heavy density (6S) were determined by density-gradient centrifugation (25). The rate of disappearance of normal density Re provides the measure of Re degradation, and the rate of appearance of heavy density Re represents Re synthesis.

time of incubation, and this represents the preexisting receptor being degraded under these experimental conditions. Comparing cells exposed to E alone and E + P, at 1 h there is no difference in the Re profile, but by 3 h, both Re peaks are substantially reduced with E + P. By 6 h, the P effect is clear, and by 9 h, the effect is rather pronounced. From the loss of the 4S normal density peak, we can calculate Re turnover, and within 3 h, P causes a significant enhancement of the Re turnover (Fig. 7). The data from the second peak, when corrected for the difference in receptor turnover, provide a measure of Re synthesis (25). With E alone, the rate of Re synthesis increases by 9 h, when there is a significant difference in Re synthesis, comparing E with E + P (Fig. 7). What is the interpretation of these results? First, there is a rapid effect of P on Re turnover, within 3 h, and this is maintained throughout the period studied. Under these culture conditions, P has no effect on Re synthesis, and the rate of Re synthesis doesn't change with E + P (Fig. 7). What P appears to do is block E-induced Re synthesis, which is the so-called replenishment response that one sees with E action (Fig. 1). However, this happens subsequent to the P-induced change in Re turnover. On the basis of this kind of evidence, we propose the first effect of P is on receptor degradation and there is a secondary, later response on Re synthesis that may be a consequence of the earlier effect on Re turnover (26).

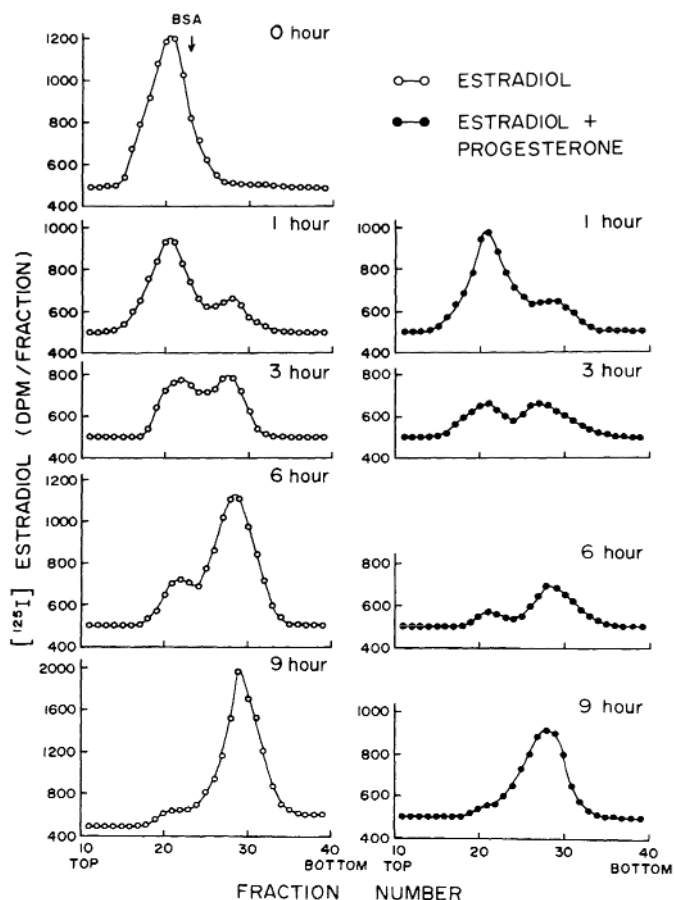


Fig. 6. Sedimentation profiles of nuclear Re as determined by the density shift technique. Decidual cell cultures were incubated with either 1 nM estradiol (open circles) or 1 nM estradiol plus 100 nM progesterone (closed circles) in the presence of heavy amino acids as depicted in Fig. 5. After 1, 3, 6, and 9 h of incubation, nuclear Re was extracted and labeled with [125 I]-estradiol, and the heavy and normal density forms of Re were separated by density-gradient centrifugation (25). Bovine serum albumin (BSA) was used as a sedimentation marker (4.6S).

How can we explain P-induced Re degradation? Since we have not been able to show that P induces a factor that directly acts on the receptor, there must be something else involved. The missing link appears to be the nuclear acceptor sites for Re. Fig. 8 shows acceptor site binding curves as determined with techniques that we have developed, using uterine chromatin derived from the proestrous hamster uterus. The proestrous uterus is an estrogen-dominated tissue, and the chromatin is prepared and coupled to cellulose for binding to labeled ReE complex that is also derived from the proestrous uterus. If we treat the proestrous animal for 2 h with P *in vivo*, and then harvest the chromatin in the same way, and perform a binding assay with the same labeled ReE preparation, we find a substantial decrease in the binding of hormone-receptor complex to chromatin (Fig. 8). This difference in the binding capacity of control and P-treated chromatin can be accounted for by specific saturable binding sites since they can be competed with unlabeled ReE complex (Fig. 8). A similar loss of Re acceptor

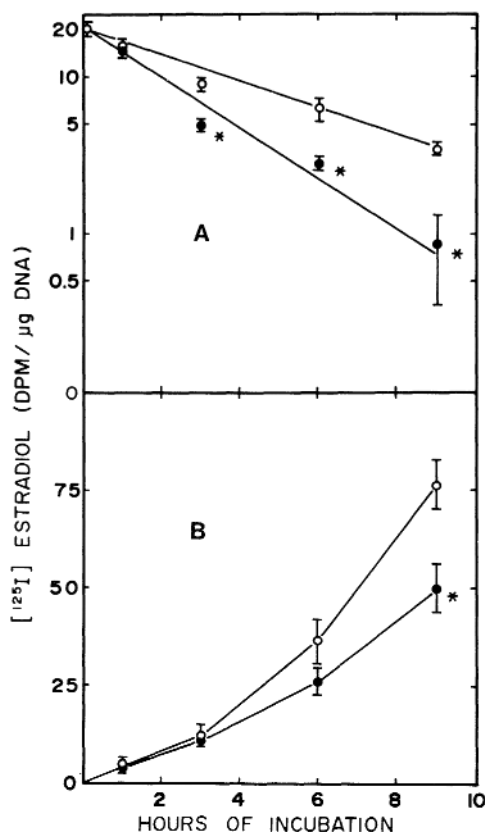


Fig. 7. Estrogen receptor (Re) degradation (panel A) and synthesis (panel B) as influenced by progesterone action in decidual cells. The half life ($t_{1/2}$) of nuclear Re was determined from the regression lines calculated from the normal density (4S) Re peak (see Fig. 6). The $t_{1/2}$ of nuclear Re was 3.7 h for cells incubated with E₂ (open circles) and P treatment (E₂ + P, closed circles) shortened the $t_{1/2}$ of Re to 1.9 h. Synthesis of Re was calculated from the size of the heavy density (6S) peak using the formula: $K_s = K_d \cdot Rt / (1 - e^{-K_d t})$, where K_s is the synthesis rate, $K_d = 0.693/t_{1/2}$, and Rt is the Re value (DPM/μg DNA) at time "t". With this approach, the corrected synthesis ($K_s \cdot t$) of nuclear Re was calculated and plotted in panel B. The Re synthesis rate (K_s) averaged 4.6 DPM/μg DNA/h for E₂ plus P treatment, whereas E₂ treatment alone increased the Re synthesis rate from 4.6 DPM/μg DNA/h at 1 h to 7.6 DPM/μg DNA/h at 9 h of incubation. From Leavitt and Takeda (25).

sites is observed on the day of proestrus, before and after the endogenous preovulatory P surge, as compared to 2 h and 4 h of P treatment before the time of the P surge (Fig. 9). The Re-acceptor binding results suggest that P acts either at the level of the ReE complex as it exists on acceptor sites or on the acceptor site itself (Fig. 10).

We have attempted to determine whether P-dependent modification of the receptor protein could account for decreased nuclear retention. Ovariectomized hamsters were treated with E and P implants for 15 days. This hormonal regimen is similar to that which occurs during the 16-day gestation period of the hamster. On day 15, P implants were removed from one group and left in the other group, and the E implants remained in both groups. Activated cytosolic ReE was prepared from both groups and then incubated with proestrous uterine hamster chromatin to determine if dif-

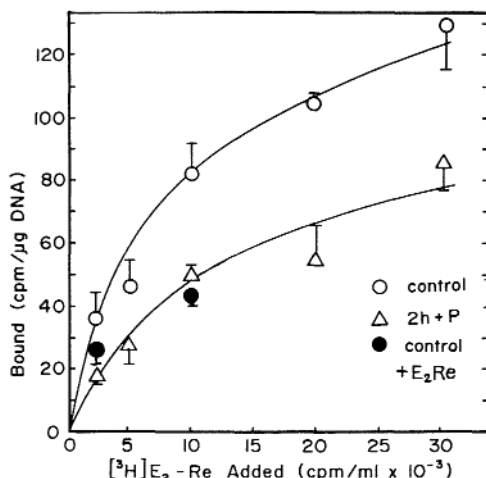


Fig. 8. Progesterone (P)-induced depletion of nuclear Re acceptor sites from uterine chromatin. Proestrous hamsters were treated for 2 h with 5mg P (triangles) or oil vehicle (control, open circles). Chromatin was isolated and linked to cellulose as described by Spelsberg et al. (10). Then, binding assays were performed at 0° C using different concentrations of [³H]E₂-Re complex prepared from uterine cytosol (obtained from control animals) and concentrated by ammonium sulfate (45%) precipitation prior to assay. A competition experiment was done using a 5-fold excess of radioinert E₂-Re complex (closed circles).

ferences existed in the ability of these receptor preparations to bind to Re acceptor sites. The saturation profile of P-withdrawn ReE was identical to that of P-treated ReE, indicating no effect on the receptor protein. In contrast, P action *in vivo* significantly decreases nuclear retention of ReE. We conclude from these studies that the receptor itself is not the substrate for the P-induced activity (factor), and that nuclear ReE loss must occur by an alternative mechanism at the level of the chromatin acceptor sites (Fig. 10).

DISCUSSION

We have demonstrated that P selectively down regulates the occupied form of nuclear Re, and there is a rapid depletion of nuclear acceptor sites within 2 h. We found that P stimulates Re turnover within 3 h, and subsequently P inhibits E-induced Re synthesis at 9 h. Thus, we propose that it is the P-induced depletion of nuclear acceptor sites which appears to enhance nuclear Re turnover (Fig. 10) and thus interrupt at least some aspects of E-dependent gene transcription. Our working hypothesis is that P somehow blocks or alters the high-affinity acceptor sites for Re, and

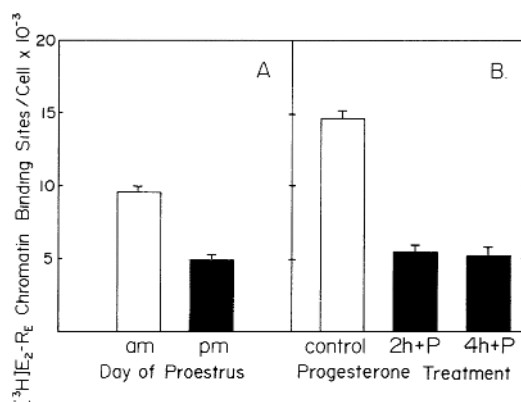


Fig. 9. Nuclear Re acceptor sites in hamster uterine chromatin as a function of endogenous P secretion (panel A) or treatment with exogenous P (panel B). On the day of proestrus, animals were studied at 10 am before the preovulatory P surge (see Fig.2) and at 7 pm during the P surge. P treatment (5mg/animal) or oil vehicle (control) was given at 8 am on the day of proestrus, and animals were killed 2 h and 4 h later. The acceptor site assays were conducted as shown in Fig. 8. The total number of acceptor sites was determined by Scatchard plot analysis of the Re binding data, and the sites per cell was calculated assuming 6 pg DNA per cell.

this, then, leads to an enhancement of Re turnover. This model would require a rather rapid mechanism for degrading receptor, at least as we recognize it from steroid binding studies, and P appears to increase the half-time of Re turnover from 4 h to 2 h. Thus, our studies show that P doubles the Re degradation rate by some mechanism that stimulates the already rapid turnover ($t_{1/2} = 4h$) of the Re protein in the target cell nucleus. The amino-terminal amino acid composition of a protein appears to determine the protein's half-life (27), and proteins with PEST regions rich in proline (P), glutamic acid (E), serine (S) and threonine (T) are rapidly

degraded in eukaryotic cells (28). Recent information on the amino acid sequence of the human Re (29,30) supports the idea that the rapid turnover of nuclear Re could be related to the amino acid composition of one or more regions of the receptor protein. Exactly how the P-induced loss of nuclear acceptor sites relates to Re degradation remains to be determined.

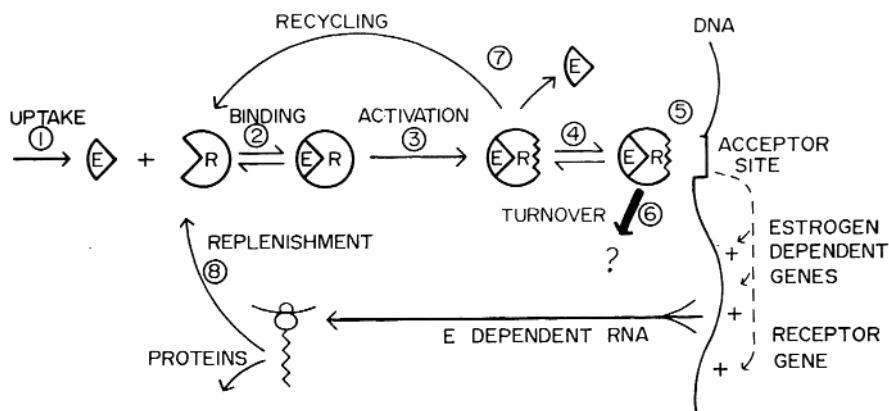


Fig. 10. Proposed site of P action in the down regulation of the Re system. Results are reviewed (see text) showing that P has no effect on E uptake (step 1), E binding to Re (step 2), Re activation (step 3), or the ability of ReE complex to bind to acceptor sites (step 4). We demonstrated that P causes a rapid (within 2 h) depletion of Re acceptor sites, thus preventing the binding of ReE complex to chromatin (step 5); this appears to lead to increased turnover of ReE (step 6) within 3 h followed by a decreased E-dependent gene transcription and reduced levels of E-dependent proteins such as Rp, oxytocin receptor and the Re protein (step 8, Re replenishment). The mechanism regulating the rapid degradation of Re protein (step 6) is not well understood, and our evidence suggests that a P-induced factor (ReRF) acts on step 5 rather than step 6.

Initial studies of P and E interaction demonstrated a P-dependent decrease in cellular ReE levels but did not identify which Re form was affected (31). Such knowledge was vital in formulating a mechanistic hypothesis; thus, we have addressed the question of selective regulation of different Re populations by P. In several different systems, including the sheep uterus (32), rodent uterus (19,33), decidual cell cultures (25,34), and T47-D (clone 11) human mammary carcinoma cells (35), we found that P caused a rapid and selective decrease in nuclear Re. The later effect of P on cytosolic Re titer occurs after 6 h, and appears to result from the earlier P effect on nuclear Re (19,21), although a separate control of cytosol Re may exist under conditions of prolonged P action (36).

Recent results from other laboratories have confirmed our findings. Investigators have demonstrated E- or P-dependent down-regulation of nuclear Re in pituitary (37,38), hypothalamus (39), and other regions of the brain (40) and the reproductive tract (41). That P-dependent nuclear ReE loss is indeed an important regulatory event is supported by the subsequent loss of E-dependent gene expression. The connection between nuclear ReE retention and E-stimulated gene transcription is well established (42-44). Although we have demonstrated a role of P in down regulating nuclear ReE in several systems, we need to understand more precisely how this alters estrogen action. Toward that end, we have examined the effect of P withdrawal on nuclear Re recovery and the subsequent response of E-dependent oxytocin receptor, Rp, and cytosolic Re (23,32,36). A significant correlation exists between nuclear Re recovery and the synthesis of E-dependent proteins. Thus, P-dependent down regulation of the nuclear ReE may be coupled with a decreased rate of synthesis of several E-dependent proteins.

Having established a case for P modulation of E action by down regulation of nuclear ReE, we need to generate more information about the mechanism that is involved. A number of investigators have recently studied Re turnover and we will examine their findings, compare them with our findings, and attempt to identify the factors which play a role in Re regulation.

Several workers have compared the effects of various estrogen agonists and antagonists (3,45,46), elevated temperature (47), protease and phosphatase inhibitors (48,49), and inhibitors of protein and mRNA synthesis on ReE turnover (22,50). Density shift experiments (25,26,34) and chromatin binding studies were also done (50).

The importance of receptor occupancy in decreasing the half-life of Re is suggested by several studies. The occupied or ligand-associated form of nuclear Re seems to be the most susceptible to P action (20,21). In non-P-treated cells, Eckert et al. (46) showed by the density shift technique that unoccupied Re had a half life of 4 h while occupied Re had a half life of about 3 h. When the estrogen antagonist, nafoxidine, was used as the ligand, the half time of the Re-nafoxidine complex was 3.5 h. Earlier work by Horwitz and McGuire (51,52), using either estrogen agonists or antagonists, reported a rapid and progressive depletion of nuclear Re following the addition of estrogen to the medium of cultured MCF-7 cells. Antiestrogens failed to deplete nuclear Re. These authors postulated that antiestrogen binding impeded receptor "processing" and that processing was a prerequisite to E-dependent gene expression. They proposed a model in which Re processing was necessary for hormone action, but this model can be questioned on the basis that the apparent lack of receptor processing with antiestrogen is due to the slow clearance of these agents from the nucleus (53,54). Although it is possible that receptor which is occupied by agonist (E) may be somewhat more susceptible to proteolysis than unoccupied receptor or receptor filled with antagonist, the inability of antiestrogens to inhibit the Re turnover rate indicates that receptor occupancy is not required for rapid receptor turnover. It would appear, therefore, that the activated form of nuclear Re is readily accessible to either specific or nonspecific receptor proteases in the target cell nucleus.

A second characteristic of P-enhanced nuclear ReE turnover is its dependence on elevated temperature (47-49). A series of experiments conducted in our laboratory examined differences in the amount of nuclear ReE recovered from P-treated or control nuclei which had been incubated at either 0° C or elevated temperature (20-36° C). At low temperatures, the amount of nuclear ReE extracted by KCl, thiocyanate, or pyridoxal 5'phosphate from control or P-treated nuclei was the same. However, at elevated temperature, a discrepancy became apparent; a significantly larger amount of nuclear ReE

was recovered from the nuclei of untreated cells than from the nuclei of P-treated cells. Increased turnover rate with elevated temperature is characteristic of enzyme-catalyzed reactions; thus, it was proposed that a P-activated factor or activity was involved in receptor degradation (2). Additional evidence from our laboratory supported the hypothesis that loss of nuclear ReE required enzymatic activity. It was noted that the temperature dependent, P-induced loss did not occur if isolated nuclei from P-treated target tissue were prepared in buffer containing phosphate, molybdate, or vanadate (47-49). Since the last three reagents are acid phosphatase inhibitors, it seemed possible that the enzymatic factor was a phosphatase whose dephosphorylation of the nuclear Re perhaps altered Re half life. However, it was also possible that agents such as molybdate interact directly with the Re protein instead of the phosphatase, since several studies have demonstrated the ability of molybdate to stabilize steroid hormone receptors (55-57).

If P-action requires a specific enzyme activity (e.g., ReRF), pretreatment with puromycin, cycloheximide and actinomycin D to block its synthesis prior to P-treatment might also block subsequent nuclear ReE loss. Such inhibition was demonstrated when inhibitors were administered in vitro to uterine strips 30 min prior to the addition of P (22). Horwitz and McGuire (52) reported that actinomycin D was capable of inhibiting nuclear ReE turnover in E-treated MCF-7 cells. The inhibition of Re turnover was attributed to the GC specific intercalating properties of actinomycin D, suggesting that altered DNA conformation inhibited receptor processing by some unknown mechanism. Neither AT specific nor nonspecific DNA intercalators nor cycloheximide inhibited nuclear ReE turnover rates as effectively as did actinomycin D. The apparent inconsistency between those findings and ours may reflect differences in the mechanism of P-dependent vs. E-dependent nuclear Re processing.

It is important to emphasize that in all studies to date, nuclear ReE levels were measured using labeled-steroid binding assays. As such, these assays are designed to detect only those receptor forms capable of binding ligand. With this in mind, the possibility had to be considered that the apparent loss of nuclear Re by P action may affect only ligand-Re binding while leaving the remainder of the receptor intact. Decreased ligand binding could be caused either by interaction of another protein with the receptor, or by changes in the conformation of the receptor or by covalent modification of the receptor protein (58). Since nuclear ReE complexes from P-treated cells have no detectable physicochemical differences as compared to those from control cells (2), such modification of Re would have to be a subtle one, perhaps involving phosphorylation of a single key amino acid residue.

Work of Auricchio and colleagues (59-62) demonstrated that an Re tyrosine residue is a substrate for endogenous nuclear phosphatase and cytosolic kinase action. Tyrosine phosphorylation has been implicated as a key regulatory event in cellular differentiation and growth. Phosphorylation of Re tyrosine residues enhances Re estrogen binding capacity without altering Re affinity for ligand. These investigators observed an apparent "inactivation" or loss of E-binding capacity which was attributable to dephosphorylation of receptor tyrosine residue(s). They proposed that, in vivo, a nuclear phosphatase caused an apparent loss of Re by inhibiting ligand-receptor interaction. A potential involvement of phosphorylation in Re regulation was also demonstrated by Moudgil and Eessalu (63). In their studies, incubation of cytosol ReE with ATP at 0-4° C for 1 h enhanced ReE binding to nuclei and DNA-cellulose. When a nonhydrolyzable ATP analog was substituted for ATP in the incubation, no enhanced activation was observed. Similarly, Fleming et al. (64) showed that cGMP, ATP and GTP, but

not cAMP, increased the apparent recovery of ReE from nuclei. They proposed that a cGMP-dependent protein kinase was instrumental in Re activation. Unfortunately, no effect of receptor dephosphorylation on DNA-Re interactions has been demonstrated.

Although binding assays are a useful approach to the study of ReE turnover, they provide little information about putative non-ligand binding forms of Re. Monsma et al. (65) used tamoxifen aziridine to covalently label the receptor and follow its degradation. The loss of covalently-bound ligand was assumed to parallel loss of Re. Surprisingly, the loss of the ligand binding site and the loss of immunoreactive Re protein occurred simultaneously (65). This suggests that significant changes in the receptor's tertiary structure accompany degradation or inactivation of the functional ligand binding site. On this basis, antibodies to receptor may prove very useful in the detection of receptor products formed during Re processing in the nucleus. Certainly, monoclonal antibodies to conformation-dependent epitopes will provide information about how receptor structure changes under various experimental conditions (64).

Recent studies indicate that quaternary changes in receptor structure may be even more significant than tertiary changes in regulating Re function. Immunocytochemical analyses of receptor forms occurring during short-term (30 min) or longer-term (6 h) E exposure revealed only one form of nuclear receptor in short-term E-treated cells (66,67). This Re form was an activated monomer (4S) showing rapid ligand dissociation (dissociation rate = 8.5 min). After longer exposure to E, nuclei contained both the monomeric form and an activated 8.8S dimer. The dimer elutes differently from hydroxylapatite and has a slower dissociation rate (216 min) compared to the monomeric form. There doesn't appear to be a differential turnover of different Re forms in MCF-7 cells. Monsma et al. (65) allowed cells to incorporate heavy amino acids (^{13}C , ^{15}N , ^3H) into Re which was first charged with ligand and then covalently crosslinked to adjacent macromolecules. Only one major receptor form, a 130,000 da homodimer, was evident, and the subunits of the homodimer were degraded simultaneously. Gorski's (68) and Notides' (69) laboratories have provided additional information regarding monomer-dimer interactions, and both groups present evidence for dimer interactions, although they disagree on the sequence of events involved and the significance of receptor dimerization to hormone action.

A Unifying Concept

We have been unable to detect physicochemical modifications in the nuclear Re which would inhibit its retention and accelerate its turnover rate (2). Therefore, it is likely that the P-induced decrease in nuclear ReE retention results from a change in the nuclear binding site for receptor rather than a change in the receptor itself. If the nuclear Re binding site were lost, receptor might be more susceptible to proteolysis.

Our acceptor site assays using activated ReE and 350 mM salt-washed chromatin prepared from proestrous hamster uteri are a modification of the method of Ruh and Spelsberg (70). Within 2 h or P treatment *in vivo*, the ReE binding capacity of crude chromatin is markedly decreased with no P-dependent change in binding affinity for receptor. This P effect is not inhibited by cycloheximide at 2 h after P, but it is completely blocked by cycloheximide at 4 h after P (unpublished observations). This implies that P-dependent loss of Re-chromatin binding sites is a rapid and reversible process requiring replenishment of a short-lived factor. A similar effect of cycloheximide in blocking estrogen receptor turnover has been observed by Campen and Gorski (71).

The effect of P in depleting Re binding sites in chromatin is the first evidence for hormone-dependent changes in acceptor site capacity. The idea that P modulates nuclear ReE-chromatin interactions is consistent with previous reports that steroid hormones cause numerous changes in various chromatin proteins, including histone acetylation (71-75), induction and synthesis of nonhistone acidic chromosomal proteins (76,77), increased DNase I sensitivity (78), and other changes in chromatin conformation (79).

In order to consider how P action might alter the nuclear Re binding site, it is first necessary to understand the characteristics of that site. Proposed nuclear ReE binding sites include the nuclear matrix (80), DNA (81-88), basic nonhistone chromatin proteins (89), acidic nonhistone chromatin proteins (90-96), and DNA-protein complexes called "acceptor sites" (10).

Because ReE possesses the ability to activate and develop an increased affinity for DNA, earlier investigators proposed that the nuclear binding site for the receptor was DNA. This hypothesis has been tested by several investigators who used recombinant DNA technology to clone E-dependent genes (97-100). Fragments of the cloned genes were tested for a role in E-dependent changes in transcription. Elegant experiments have utilized the chicken vitellogenin II (VTG II) gene, ovalbumin gene, and conalbumin gene with their respective 5' and 3' flanking regions to construct an elaborate model of E- and P-dependent expression (79). Although a detailed account of this gene regulatory structure is beyond the scope of this discussion, we will consider the enhancer and promoter regions and their putative role in ReE binding (79). An illustration of their spatial relationship is provided in Fig. 11. From 3' (right) to 5' (left), the entire gene and its regulatory elements consist of a 3' flanking region, the transcript and its intervening sequences, a capsite or "start" site proximal to the transcript, a 30 base pair "spacer," the TATA box, a 70-90 base pair "spacer," a polymorphic GC rich sequence, another "spacer" region of greatly varying length, an "enhancer" sequence, and finally the remainder of the 5' flanking region. The enhancer sequences are 76 base-pair repeats and are located not only in the 5' flanking region, but also within the 3' flanking region, in the intervening sequences, and proximal (110 bp from the capsite) and distal (several hundred to several thousand bp from the capsite) to the regulated gene. Each of these regions may contain specific sequences to which regulatory components of the transcriptional machinery bind. The ReE may be a part of that machinery, and several of these sequences have been implicated as the receptor binding site. Jost et al. (97) demonstrated the ability of CGGP sequences to compete with genomic DNA for ReE binding. This sequence also competed with a proposed enhancer sequence approximately 600 base pairs upstream from the cloned CVT II gene. Subsequent studies demonstrated that inversion of this competing sequence within the 5' flanking region resulted in loss of E-stimulated gene transcription (98). Consistent with the assignment of this region's proximity to the ReE binding sequence is the finding that, during E treatment, this region is hypersensitive to DNase digestion. Actively transcribed regions are similarly hypersensitive, and E treatment results in increased DNase susceptibility of target tissue chromatin (101,102). Geier et al. (103) demonstrated that a 5.6S, 163 Kda form, of nuclear ReE was released by extensive DNase I digestion of MCF-7 nuclei, and Thomas and Leung (104) obtained a 7S ReE form by DNase digestion of the chromatin-receptor complex. The size observed for the ReE complex argues for something other than a simple monomeric receptor-DNA interaction. It more likely consists of receptor, DNA and certain chromatin proteins. Such a complex has been characterized for chicken oviduct (90,91), for mammalian target tissue (92,93), including the hamster uterus (50), and the receptor binding site is generally referred to as the acceptor site.

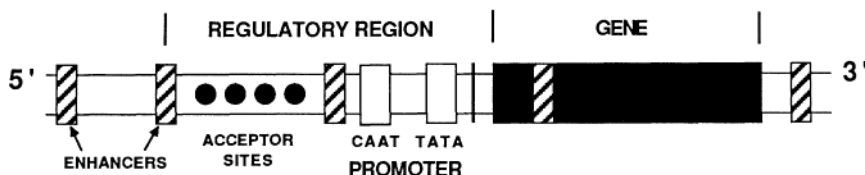


Fig. 11. Structural organization of hormone-regulated gene. The 5' flanking region contains regulatory elements such as promoter sequences (CAAT, TATA, CAP) and other sequences which recognize receptor and acceptor site proteins. Enhancer sequences appear to occur at various locations in the 5' flanking region, in the 3' flanking region, and internally in the structural gene. Acceptor proteins are proposed to bind to specific sequences in the regulatory region (adjacent to enhancers?) and receptor binding to the acceptor sites is proposed to cause looping of the DNA which appears to be necessary for active gene transcription.

The term "acceptor site" has been used to describe the putative chromatin binding site for ReE in chick oviduct nuclei (90). The acceptor appears to contain one or more acidic nonhistone chromosomal proteins which bind ReE in a specific, saturable, and high affinity manner. The acceptor protein itself can be associated with specific DNA sequences (91). The acceptor proteins can be extracted from chromatin by guanidine hydrochloride and reconstituted by gradient dialysis of the extracts in the presence of homologous DNA. Recent work from the Ruhs' laboratory suggests that the acceptor site for ReE complex in uterine chromatin can be extracted from chromatin at different guanidine hydrochloride concentrations (91-93). The data of Cushing et al. (95) indicate that there are more tissue-specific binding sites for the estrogen-Re complex (28,000 sites/cell) than for antiestrogen-Re complex (17,800 sites/cell), but both of these sites have the same affinity for receptor-ligand complex (1-2 nM Kd). These results suggest the existence of different populations of acceptor proteins which may be regulated in various ways. An alternative mechanism for regulating nuclear ReE binding to acceptor sites has been suggested by Spelsberg, who proposes the involvement of masking proteins (10). Recent work by his group provides evidence for a heterogeneous high molecular weight group of proteins which bind to the acceptor protein on or near the site which binds ReE. Because acceptor sites are demonstrable in all tissues, but masking proteins are not, the latter may be important regulators of tissue-specific gene expression.

A New Model

A model in which P-dependent loss of chromatin binding sites for ReE is the regulatory step for modulating estrogen action should satisfy several criteria. First, the receptor-acceptor interaction must be a "rate-limiting" step. Next, the modulation of this rate-limiting step must be rapid, reversible, and tightly controlled. We have incorporated recent findings

from other investigators as well as from our own studies to formulate a new model for P attenuation of E action in the mammalian uterus.

In this model, P modulates E action by decreasing the interaction of ReE with specific chromatin acceptor sites responsible for enhancement of E-dependent gene transcription. A P-dependent modification of the acceptor protein is a likely cause for such modulation, rather than a P-dependent modification of the Re itself which has not been observed to date. What type of acceptor protein modification is predicted? A number of possibilities exist, such as acceptor protein acetylation, glycosylation, phosphorylation, etc. However, the inhibitory effect of molybdate and vanadate (acid phosphatase inhibitors) on P-dependent nuclear ReE loss suggests that dephosphorylation of a key residue of the acceptor protein may prevent ReE from binding. The decreased number of chromatin binding sites would result in the observed loss of nuclear ReE retention.

How would loss of acceptor sites enhance nuclear ReE turnover? It is proposed that the non-chromatin associated ReE complex is more susceptible to either general or specific proteolysis (27,28,105) than its chromatin-bound counterpart. The receptor-acceptor complex in intact nuclei may be as large as 12S. Such an assembly of proteins may sequester and stabilize ReE, inhibiting access of proteases. Evidence for a large, tightly-bound, and a smaller, loosely-bound nuclear ReE form has been presented (67). The large form may be protected from proteolysis and the smaller form more rapidly degraded. Support for large and small nuclear Re has been provided by Muller et al. (57). When nuclear ReE of MCF-7 cells was chemically crosslinked in situ and eluted with KCl, two species were evident, a 65 Kda monomer and a 130 Kda dimer. Formation of the dimer, the effector, may be facilitated by the acceptor proteins, and this may trigger the observed increase in transcription rate. It is conceivable that P may modify the allosteric function of the acceptor protein so that formation of the receptor dimer is impeded.

How could Re dimerization stimulate gene transcription, and how could its absence diminish transcription rates? Electron micrographs of E-stimulated chick oviduct chromatin show that the chromatin is arranged in multiple "loops" with attachment points near the chromosomal matrix. These loops are DNase I hypersensitive, contain nascent RNA transcripts, and do not appear in the E withdrawn state. The attachment sites of these loops to the nuclear matrix contain a number of transcriptionally-related proteins such as topoisomerases (106), and these may contain steroid hormone receptor binding sites. A functional model for loop formation has been proposed by O'Malley et al. (43). The basis of their model was the finding that actively transcribed genes existed as "loops," and that the loops were intimately associated with the nuclear matrix. They proposed that within the loops, each E-dependent gene contained a consensus sequence which, when associated with ReE, stimulates transcription. The loop was then specifically attached to proteins of the nuclear matrix for continued stimulation of gene transcription. There would be a relationship of receptor dimers to loop conformations, and P-induced ReRF could disrupt such an array. In the absence of E, the level of nuclear ReE is low and receptor monomer formation is favored. Receptor monomers may be inhibitory or inactive in the transcription process. A simple P-dependent event such as dephosphorylation of an acceptor protein could trigger a cascade which would produce significant changes in overall chromatin conformation and gene transcription. It may be pertinent that Rp is associated with a 90 Kda protein perhaps containing protein kinase activity, although it is equally likely that Rp may activate or modulate another factor(s) responsible for acceptor protein modification (107-109). As the Rp rapidly turns over, modification of acceptor proteins ceases, and the original conformation of the nuclear ReE binding site is restored by synthesis of new

acceptor proteins, or perhaps by some reverse reaction (phosphorylation?). Thus, the model allows for reversibility and recovery of P-dependent down regulation of nuclear ReE. Of course, one should keep in mind that the down-regulated condition may be the normal or resting state, and not an inhibited state.

In summary, we have observed that P acts, via its nuclear receptor, to decrease rapidly the level of occupied nuclear Re, to increase the rate of nuclear ReE turnover, and to rapidly reduce the number of nuclear ReE chromatin binding sites. We propose that P activates a factor (ReRF) which, in turn, modifies the acceptor protein within the ReE chromatin binding site. Such modification impedes receptor association and may prevent conformational changes in chromatin required to form transcriptionally active chromatin loops. Because non-chromatin bound nuclear ReE is not afforded "protection" by its association with other chromatin components, it is more susceptible to general or specific proteolysis and its turnover rate increases two-fold. As nuclear RpP is rapidly depleted, the modification of the acceptor sites is reversed or the acceptor site is replaced, and nuclear ReE chromatin interactions are restored. Although we have advanced this model from the available evidence, it remains to be tested experimentally and must be modified as new information is generated from future experiments in this exciting field of research. New approaches using new probes for the Re protein (4) and the receptor genes (100) will undoubtedly provide new knowledge about the mechanisms controlling receptor site availability.

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Dr. Andrea Cobb (Texas Tech): The focus of my studies has been to try to identify the substrate for progesterone action and, as Dr. Leavitt discussed, we saw an effect on half-life of the estrogen receptor, and we also saw an in vivo effect of progesterone on the estrogen receptor acceptor sites. What we wanted to do was to be able to identify substrate so that we could isolate and characterize the progesterone-dependent activity which Dr. Leavitt has referred to as ReRF. We proposed two possible substrates for the ReRF and the first one we proposed was the receptor-acceptor complex. We tested this and I won't go into this in detail because we find that it did not appear to be a substrate for progesterone action. Because we had seen the in vivo evidence for progesterone down regulation of acceptor site, we followed through on the second possible substrate, which was the unoccupied or uncharged estrogen receptor-acceptor site itself. We did mixing experiments where we took the uncharged acceptor site from control chromatin and incubated it with a soluble nuclear fraction prepared from control nuclei or from nuclei that we prepared from progesterone-treated animals, and then we incubated those two mixtures at low or elevated temperature, washed it to remove any loosely-associated proteins, and then ran that in the binding assay, using Spelsberg's techniques to measure acceptor sites. We found about a 60 percent loss under these conditions of estrogen receptor binding capacity in the progesterone-treated chromatin as compared to control chromatin under the same conditions. This suggests, first of all, that the estrogen receptor-acceptor site in the control chromatin is substrate for a progesterone-dependent soluble nuclear factor. We wanted to characterize what some of the substrate modifications might be, so we used two approaches. The first was to look at DNAase sensitivity or susceptibility of the receptor in each of these cases, and the second was to further extend these studies with crosslinking experiments. In the first case, we took nuclei that were from control or progesterone-treated uterine preparations and treated them with DNAase I. In the control nuclei that we prepared, the receptor was very resistant to DNAase I digestion. However, in the progesterone-treated case, we could elute most of the receptor from the nuclei with DNAase I treatment. This suggests to us that, in the native conformation, the estrogen receptor and control acceptor site is bound to a DNAase I hypersensitive region and that such binding confers protection against digestion, whereas in the progesterone-treated case, protection is not conferred. We wanted to guard against possible differences in affinity between the receptor-acceptor site in the two treatments, so we crosslinked the receptor in place with formaldehyde and again proceeded with the digestion. This fixed a portion of the receptor in place with the progesterone-treated acceptor sites, but there was still a large percentage that were removed. In the last experiment, we tried to elute receptor with KCl. Using 400 mM KCl and crosslinked nuclei, in the progesterone-treated case, we were able to extract most of the receptor from the nuclei, even though it had been crosslinked. This suggests that the receptor is not tightly-associated with its acceptor sites, whereas in the control situation, it was firmly crosslinked in place and therefore was resistant to KCl extraction. To summarize, we have shown in vitro that unoccupied estrogen receptor-acceptor sites in control chromatin are substrates for progesterone-dependent nuclear activity and that the ability of KCl and DNAase I to extract receptor from progesterone-treated chromatin, but not from control chromatin, supports the hypothesis that progesterone action may change the conformation of the uterine receptor-acceptor sites, and we are pursuing this hypothesis with other approaches at the present time.

Dr. Spelsberg (Mayo Clinic): Wendell, your results are really interesting. I've never looked at any of this, so it would be interesting to see what happens in the chick, but have you tried unmasking these sites to see if they were just covered up and not depleted?

Dr. Leavitt: For the estrogen receptor binding sites, we have been collaborating with Drs. Tom and Mary Ruh, using their approach, which is to extract the chromatin with guanidine hydrochloride as you do, and unfortunately, none of these sites seem to be the ones that are down regulated by progesterone. Dr. Cobb has only been able to see this effect of progesterone on chromatin using crude chromatin. When guanidine hydrochloride is used even one molar, we lose binding and so we think it's a different population of sites than the ones that the Ruhs have reported.

Dr. Chegini (Louisville): My question has to do with the conditions that you use for your chromatin. I think the extraction of the chromatin is very important in your experiments, particularly when you treated it with DNase I and the high level of potassium chloride. You are relaxing the chromatin fibers and, as a result of that, you might argue this damage to remove some of your acceptor sites when you have different concentration of acceptor sites or the number of acceptor sites goes up and down.

Dr. Leavitt: That's very true. Our acceptor preparation is not first exposed to any of those reagents. In other words, when we measure acceptor sites using our regular assay procedure, we are very cautious to preserve the conformation of chromatin in what we hope is the native conformation. But the point of Dr. Cobb's work was that after progesterone action, the conformation of the chromatin is changed in such a way that the acceptor sites are no longer present. That is, the sensitivity to DNase is different. The KCl extractability is different, even after the crosslinking step. So, on the basis of those treatments, we are proposing that progesterone is having some effect on the conformation of chromatin and changing the acceptor sites that exist in the native state.

Dr. Stancel: I was wondering, in those experiments where you did the salt washout, do you have any indication that the effect of progesterone is specific for washout of the estrogen receptor, or are all the nuclear proteins getting washed out?

Dr. Cobb: I think that you would find the same type of proteins being released under the salt wash or DNase I as you would if you used the intact nuclei, probably less, because certainly some of them have been removed during the preparative treatments, but I certainly don't think that it would be specific for estrogen receptor per se.

Dr. Okulicz: Wendell, do you have enough data to determine whether or not the acceptor site loss occurs prior to changes in the degradation rate of receptor?

Dr. Leavitt: We were able to detect a loss of acceptor site at 2 hours and we found stimulation of receptor turnover at 3 hours. We haven't detected a loss of receptor sites in any shorter periods.

Dr. Okulicz: I was curious if you are able to determine whether or not that precedes the change in degradation of receptor.

Dr. Leavitt: We'd love to know the answer to that.

Dr. Spelsberg: Wendell, it would be interesting with your acceptor proteins for estrogen receptor to see if there would be crossreactivity with antibodies to acceptor site for progesterone receptor.

Dr. Leavitt: Indeed, it would.

Dr. Brenner: I'm always worried about progesterone action because it down regulates its own receptor, so the question is, is there some threshold level of progesterone receptor that, to which, you can be depressed and still have these effects. What is the relationship between what you've shown on the effects of progesterone on the estrogen receptor linkage and its own receptor? Does anything happen before the progesterone receptor is suppressed?

Dr. Leavitt: No, they seem to follow a very similar time course. We haven't worked this out, again, in very great detail minute by minute, because we don't have the right probes, but the progesterone receptor down regulation and the nuclear estrogen receptor down regulation seem to be following a very similar time course. Whether this is coincidental or related, I don't know. It would be lovely if the progesterone receptor were the driving force, but we haven't been able to prove that as yet.